

Joshua Aryeetey

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EDUCATION

Toronto Metropolitan University

Bachelor of Engineering, Computer Engineering (Software Specialization)

Sept. 2024 - Apr. 2028

Toronto, Ontario, Canada

- Selected Coursework: Algorithms & Data Structures, Software Systems, Database Systems I, Microprocessor Systems

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, MATLAB, HTML, CSS

Application & Backend: React Native, Expo, Node.js, Express, FastAPI, REST APIs, Prisma, Streamlit

Data & AI/ML: PostgreSQL, MongoDB, Pinecone, Pandas, NumPy, scikit-learn, RDKit, RAG, Embeddings, Vector Search

Testing & Tools: Git, GitHub, Docker, Vitest, Jest, Supertest, pytest, Playwright, Railway, Firebase Auth, Power BI, Excel, Word, Quartus

TECHNICAL EXPERIENCE

AI/ML Research Intern | Generative Molecular Modeling

Mar. 2025 - Jun. 2025

Living in Silico

Toronto, Ontario, Canada

- Converted 15,000+ molecular records into radius-2, 128-bit Morgan fingerprint feature vectors, applying data validation and feature engineering to prepare consistent inputs for downstream machine-learning generation experiments.
- Configured and evaluated an RNN-based sequence generator for 10 epochs at batch size 64, producing 500 molecular samples and comparing generation behavior across curated training subsets.
- Applied similarity-based fragment matching with fragment merging/linking to recombine compatible structures; curating experiments to 400–600 entries increased usable-candidate yield by approximately 17%.
- Compared sequence-model and similarity-driven generation workflows, documenting outputs, failure modes, and tradeoffs across 2 approaches for technical handoff and future experimentation.

Software Engineering Intern | Data Pipelines & Automation

Sep. 2025 - Nov. 2025

Stush Patties

Toronto, Ontario, Canada

- Built Python ETL for 3 distributor feeds across CSV, XLSX, and XLSB, converting incompatible sales-report structures into one canonical schema for recurring Power BI analysis in a two-person client team.
- Normalized SKUs, sales, units, regions, case packs, and reporting periods with source-aware validation; used a bounded position/cell parser for the hardest source layout to keep the delivery workflow reliable.
- Produced a unified dataset, data dictionary, and quality report that replaced repeated source-by-source cleanup with one repeatable handoff for distributor, product, retailer, and regional reporting.

PROJECTS

Food Tracker - Full-Stack Nutrition & Search Platform | React Native, TypeScript, Express, PostgreSQL, Pinecone, Docker

- Built a React Native/TypeScript nutrition platform with Express and PostgreSQL/Prisma, combining food and weight logging, barcode capture, search, analytics, recommendations, and AI-assisted input across 12,000+ normalized foods.
- Benchmarked 80 search queries and raised Top-1 retrieval from 50% to 89% (40/80 to 71/80), recovering 32 of 40 legacy misses with fuzzy retrieval plus deterministic ranking and 0 measured regressions.
- Unified 5 trusted nutrition sources behind 1 authoritative search and serving model, separating candidate retrieval from nutrition truth so ranking could improve without compromising stored nutrition values.
- Hardened the product for real use with Firebase Auth, Railway staging, health/readiness checks, 1,700+ automated API/mobile tests, and physical-iPhone validation before deployment.

Cho'Veigo - AI Job Discovery & Resume Tailoring Platform | Python, FastAPI, Streamlit, Gemini

- Co-developed a FastAPI/Streamlit career platform that combined job discovery, evidence-based matching, and resume/cover-letter generation, reducing my end-to-end application-prep workflow to roughly 3–10 minutes before external forms.
- Expanded provider-neutral job discovery across Greenhouse, Lever, and controlled first-party sources to a 41-company registry, with 10 enabled sources producing real feeds of 100+ postings at the documented milestone.
- Replaced brittle 100-point keyword scoring with evidence-authoritative fit and eligibility logic that matched manual review on 19/20 validation fit grades and 20/20 eligibility decisions.
- Hardened first-party retrieval with static-first requests, bounded Playwright fallback, retries, and source isolation so one failing career site could not break the full refresh.

Crest - AI Expense Intelligence Platform (3rd Place - Brim Financial Challenge) | Python, FastAPI, MongoDB, Gemini

- Built FastAPI/MongoDB backend workflows across 4,000+ anonymized expense transactions, combining deterministic policy checks, preapproval logic, rule-based anomaly signals, and AI-assisted financial reporting.
- Implemented policy RAG by chunking PDFs, generating 3,000+-dimensional Gemini embeddings in MongoDB Atlas, and retrieving the top 5 of 100 vector-search candidates to ground policy analysis.